Experiment 6

Aim: Experiment to implement any one of the clustering algorithms using Rapid Miner and Python.

Theory :

Clustering is an unsupervised machine learning technique used to group similar data points based on their characteristics. It helps in identifying patterns and structures within a dataset without prior labels. Common clustering algorithms include K-Means, which partitions data into k clusters, DBSCAN, which finds density-based clusters and outliers, and Hierarchical Clustering, which creates a tree-like structure of clusters. Clustering is widely used in market segmentation, anomaly detection, image analysis, and bioinformatics. The choice of clustering algorithm depends on data distribution, noise levels, and the desired cluster shapes.

Clustering is an unsupervised machine learning technique used to group similar data points together without using labelled data. It helps discover hidden patterns or natural groupings in datasets by placing similar data points into the same cluster.

- Discover the natural grouping or structure in unlabelled data without predefined categories.
- Data points are assigned to clusters based on similarity or distance measures.
- Uses Euclidean distance, cosine similarity or other metrics depending on data type and clustering method.

Types of Clustering

1. Hard Clustering

Hard clustering assigns each data point to exactly one cluster. A data point cannot belong to multiple clusters, making the grouping clear and easy to interpret.

- Each data point belongs to only one cluster
- No overlap between clusters
- Simple and easy to interpret

2. Soft Clustering

Soft clustering allows a data point to belong to multiple clusters with different probabilities. Instead of assigning a strict cluster, it gives a degree of membership to each cluster.

Example

A data point may belong 70% to Cluster 1 and 30% to Cluster 2, indicating that it shares characteristics with both groups.

Use Cases

- **Overlapping class boundaries:** Useful when data points cannot be clearly separated into distinct groups.
- **Customer personas:** Helps represent customers who share traits with multiple behavioral groups.
- **Medical diagnosis:** Patients may show symptoms related to multiple conditions.

Benefits

- **Captures ambiguity:** Represents uncertainty when cluster boundaries are not clear.
- **Models gradual transitions:** Allows smooth transitions between clusters instead of strict separation.

K-Means Clustering:

K-Means Clustering groups similar data points into clusters without needing labeled data. It is used to uncover hidden patterns when the goal is to organize data based on similarity.

- Helps identify natural groupings in unlabeled datasets
- Works by grouping points based on distance to cluster centers
- Commonly used in customer segmentation, image compression and pattern discovery
- Useful when you need structure from raw, unorganized data.

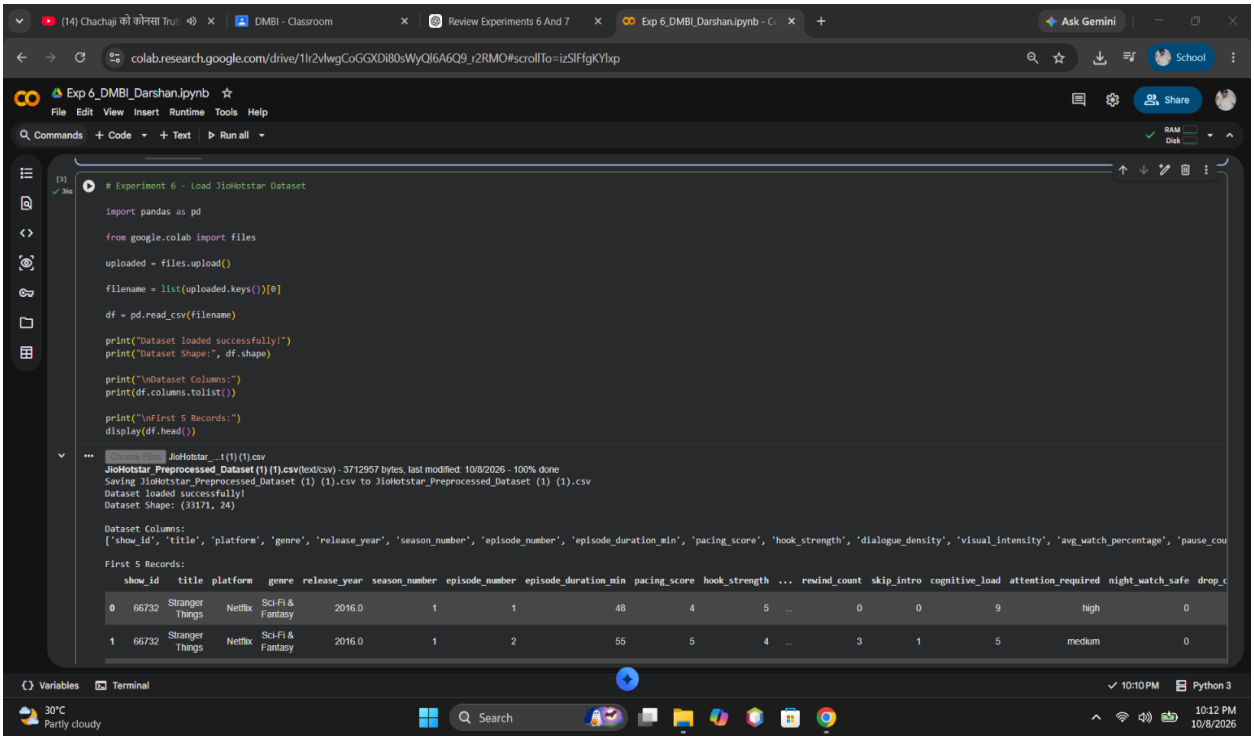
Uses of K-Means Clustering

K-Means is popular in a wide variety of applications due to its simplicity, efficiency and effectiveness. Here's why it is widely used:

1. **Data Segmentation:** One of the most common uses of K-Means is segmenting data into distinct groups. For example, businesses use K-Means to group customers based on behavior, such as purchasing patterns or website interaction.
2. **Image Compression**: K-Means can be used to reduce the complexity of images by grouping similar pixels into clusters, effectively compressing the image. This is useful for image storage and processing.
3. **Anomaly Detection:** K-Means can be applied to detect anomalies or outliers by identifying data points that do not belong to any of the clusters.
4. **Document Clustering:** In natural language processing (NLP), K-Means is used to group similar documents or articles together. It's often used in applications like recommendation systems or news categorization.
5. **Organizing Large Datasets:** When dealing with large datasets, K-Means can help in organizing the data into smaller, more manageable chunks based on similarities, improving the efficiency of data analysis.

Implementation :

A. Python



```
# Experiment 6 - Load JioHotstar Dataset

import pandas as pd

from google.colab import files

uploaded = files.upload()

filename = list(uploaded.keys())[0]

df = pd.read_csv(filename)

print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)

print("\nDataset Columns:")
print(df.columns.tolist())

print("\nFirst 5 Records:")
display(df.head())
```

```
Choose Files  JioHotstar_...t (1) (1).csv
JioHotstar_Preprocessed_Dataset (1) (1).csv(text/csv) - 3712957 bytes, last modified: 10/8/2026 - 100% done
Saving JioHotstar_Preprocessed_Dataset (1) (1).csv to JioHotstar_Preprocessed_Dataset (1) (1).csv
Dataset loaded successfully!
Dataset Shape: (33171, 24)

Dataset Columns:
['show_id', 'title', 'platform', 'genre', 'release_year', 'season_number', 'episode_number', 'episode_duration_min', 'pacing_score', 'hook_strength', 'dialogue_density', 'visual_intensity', 'avg_watch_percentage', 'pause_cou

First 5 Records:
```

	show_id	title	platform	genre	release_year	season_number	episode_number	episode_duration_min	pacing_score	hook_strength	...	rewind_count	skip_intro	cognitive_load	attention_required	night_watch_safe	drop_c
0	66732	Stranger Things	Netflix	Sci-Fi & Fantasy	2016.0	1	1	48	4	5	...	0	0	9	high	0	
1	66732	Stranger Things	Netflix	Sci-Fi & Fantasy	2016.0	1	2	55	5	4	...	3	1	5	medium	0	

Inference: The JioHotstar dataset was successfully loaded into Python using the pandas library. The dataset contains information about episode characteristics and viewer behaviour. Checking the number of rows and columns helps us understand the size and structure of the data before applying clustering.

```
# Experiment 6 - Select features for K-Means clustering

features = [
    'episode_duration_min',
    'pacing_score',
    'hook_strength',
    'visual_intensity',
    'avg_watch_percentage',
    'pause_count',
    'rewind_count',
    'skip_intro',
    'cognitive_load',
    'night_watch_safe'
]

X = df[features]

print("Selected Features:")
print(features)

print("\nShape of selected data:", X.shape)

print("\nFirst 5 rows of selected features:")
display(X.head())
```

```
Selected Features:
['episode_duration_min', 'pacing_score', 'hook_strength', 'visual_intensity', 'avg_watch_percentage', 'pause_count', 'rewind_count', 'skip_intro', 'cognitive_load', 'night_watch_safe']

Shape of selected data: (33171, 10)

First 5 rows of selected features:
```

	episode_duration_min	pacing_score	hook_strength	visual_intensity	avg_watch_percentage	pause_count	rewind_count	skip_intro	cognitive_load	night_watch_safe
0	48	4	5	5	39	3	0	0	9	0
1	55	5	4	8	55	3	3	1	5	0
2	51	4	8	7	46	4	2	0	9	0
3	50	4	7	3	50	4	1	0	7	0
4	52	4	3	4	35	3	0	1	7	0

Inference: Relevant features such as average watch percentage, episode duration, hook strength, pacing score, pause count, rewind count, and cognitive load were selected for clustering. The features help identify similarities in viewing behaviour. The target-related columns drop_off and drop_off_probability were excluded because this experiment uses unsupervised learning.

```
[5]
✓ 2s
# Experiment 6 - Standardize the selected features

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

X_scaled = pd.DataFrame(X_scaled, columns=features)

print("Standardization completed successfully!")

print("\nStandardized data shape:", X_scaled.shape)

print("\nFirst 5 rows of standardized data:")
display(X_scaled.head())
```

```
Standardization completed successfully!

Standardized data shape: (33171, 10)

First 5 rows of standardized data:
```

	episode_duration_min	pacing_score	hook_strength	visual_intensity	avg_watch_percentage	pause_count	rewind_count	skip_intro	cognitive_load	night_watch_safe
0	0.451511	-1.136141	-0.303547	-0.506150	-1.415761	-0.055129	-1.293941	-0.986614	2.120815	-0.127058
1	0.757196	-0.323397	-0.881557	0.998107	-0.143579	-0.055129	0.643583	1.013568	-0.904614	-0.127058
2	0.582519	-1.136141	1.430485	0.496688	-0.859182	0.480515	-0.002259	-0.986614	2.120815	-0.127058
3	0.538850	1.136141	0.852475	1.508989	0.541136	0.480515	0.648100	0.986614	0.608100	0.127058

Inference: The selected features were standardized using StandardScaler. This converts the features to a comparable scale, generally with a mean of approximately zero and a standard deviation of approximately one. Standardization is important because K-Means uses distance calculations, and features with larger numerical values could otherwise dominate the clustering process.

```
[6]
✓ 9s
# Experiment 6 - Elbow Method

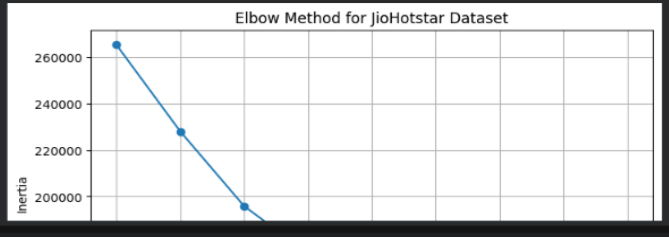
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

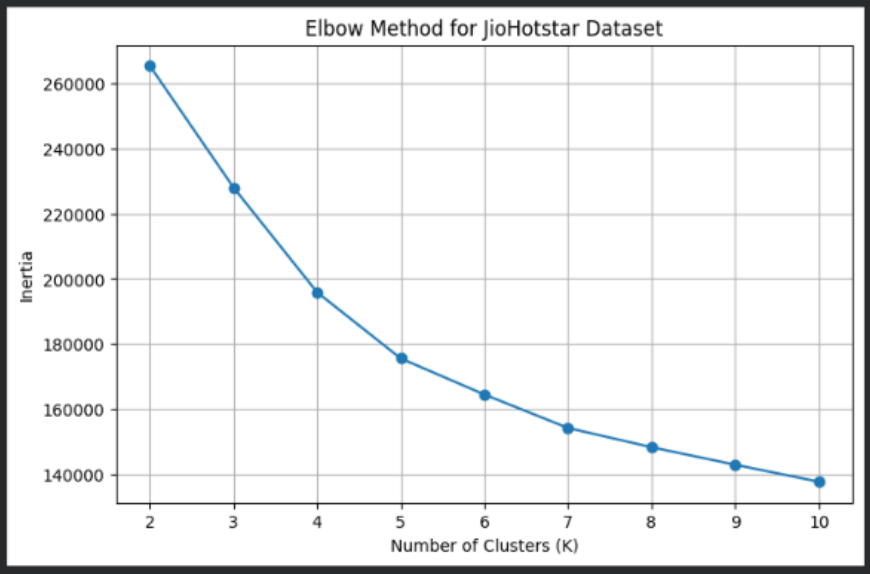
# Test different values of K
K_values = range(2, 11)

for k in K_values:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

# Plot Elbow Curve
plt.figure(figsize=(8, 5))
plt.plot(K_values, inertia, marker='o')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.title("Elbow Method for JioHotstar Dataset")
plt.xticks(K_values)
plt.grid(True)
plt.show()
```



```
plt.ylabel("Inertia")
plt.title("Elbow Method for JioHotstar Dataset")
plt.xticks(K_values)
plt.grid(True)
plt.show()
```



Inference: The Elbow Method was used to compare the inertia values for different numbers of clusters (K). Inertia measures the sum of squared distances between data points and their assigned cluster centroids. The elbow plot helps identify a suitable number of clusters. In this experiment, K = 4 was selected for clustering.

[7] ✓ 1s

```
# Experiment 6 - Apply K-Means Clustering

kmeans = KMeans(
    n_clusters=4,
    random_state=42,
    n_init=10
)

cluster_labels = kmeans.fit_predict(X_scaled)

# Add cluster labels to the original dataset
df['Cluster'] = cluster_labels

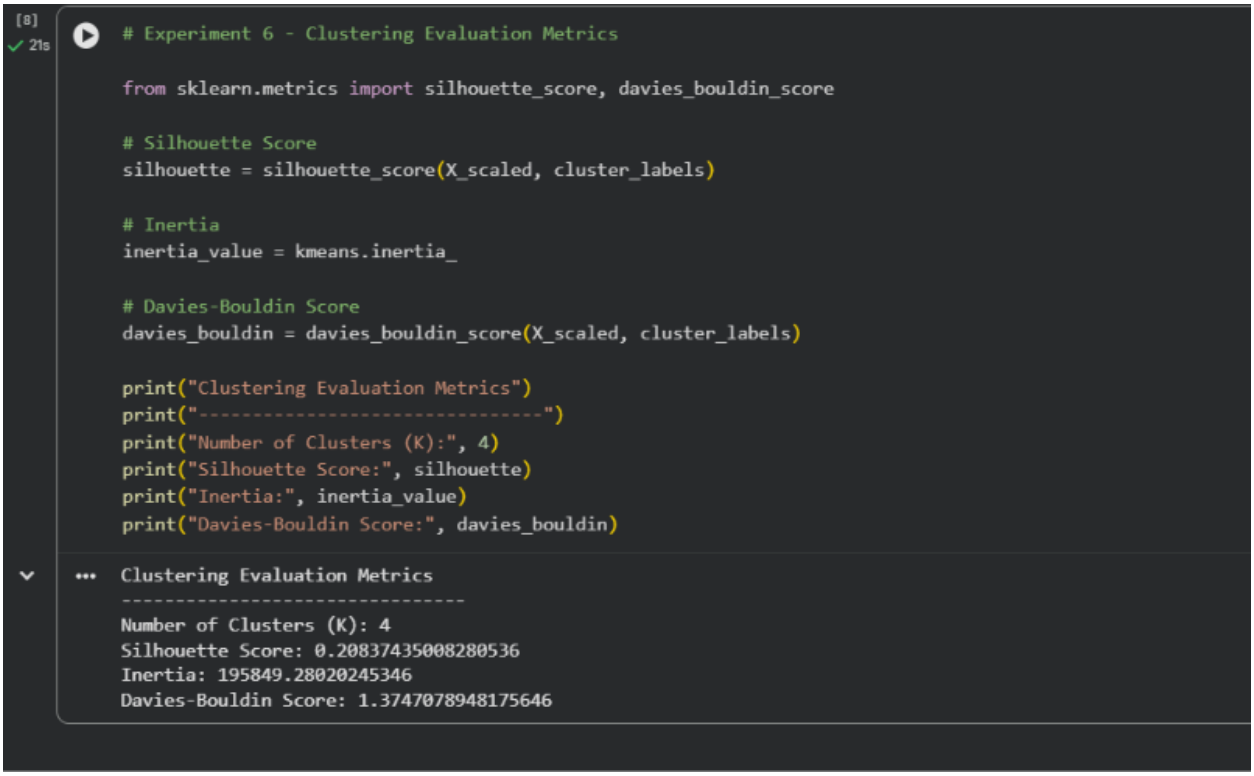
print("K-Means clustering completed successfully!")

print("\nNumber of records in each cluster:")
print(df['Cluster'].value_counts().sort_index())
```

```
K-Means clustering completed successfully!

Number of records in each cluster:
Cluster
0    13101
1    10555
2     8988
3      527
Name: count, dtype: int64
```

Inference: The K-Means algorithm was applied with four clusters using n_clusters=4, random_state=42, and n_init=10. The algorithm grouped records based on similarities in their selected features. Each record was assigned a cluster label from 0 to 3, allowing different viewer-behaviour patterns to be studied.



Inference: Clustering evaluation measures were used to assess the quality of the clusters. Inertia indicates how closely records are grouped around their centroids. The Silhouette Score evaluates how well-separated the clusters are, while the Davies–Bouldin Score measures similarity between clusters. These measures help assess the clustering result; the actual scores should be reported only if they are visible in the output.

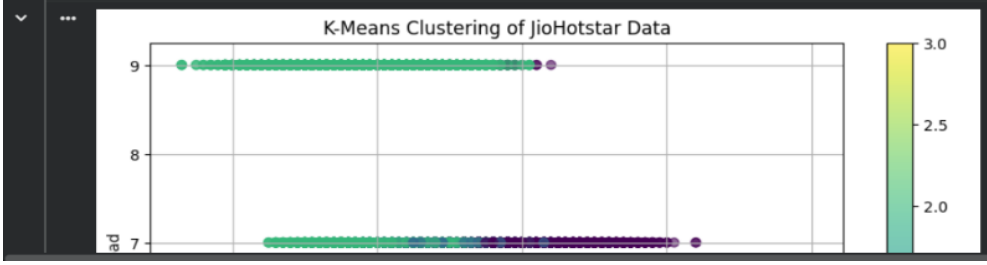
```
[9]
✓ 4s
# Experiment 6 - Visualize K-Means Clusters

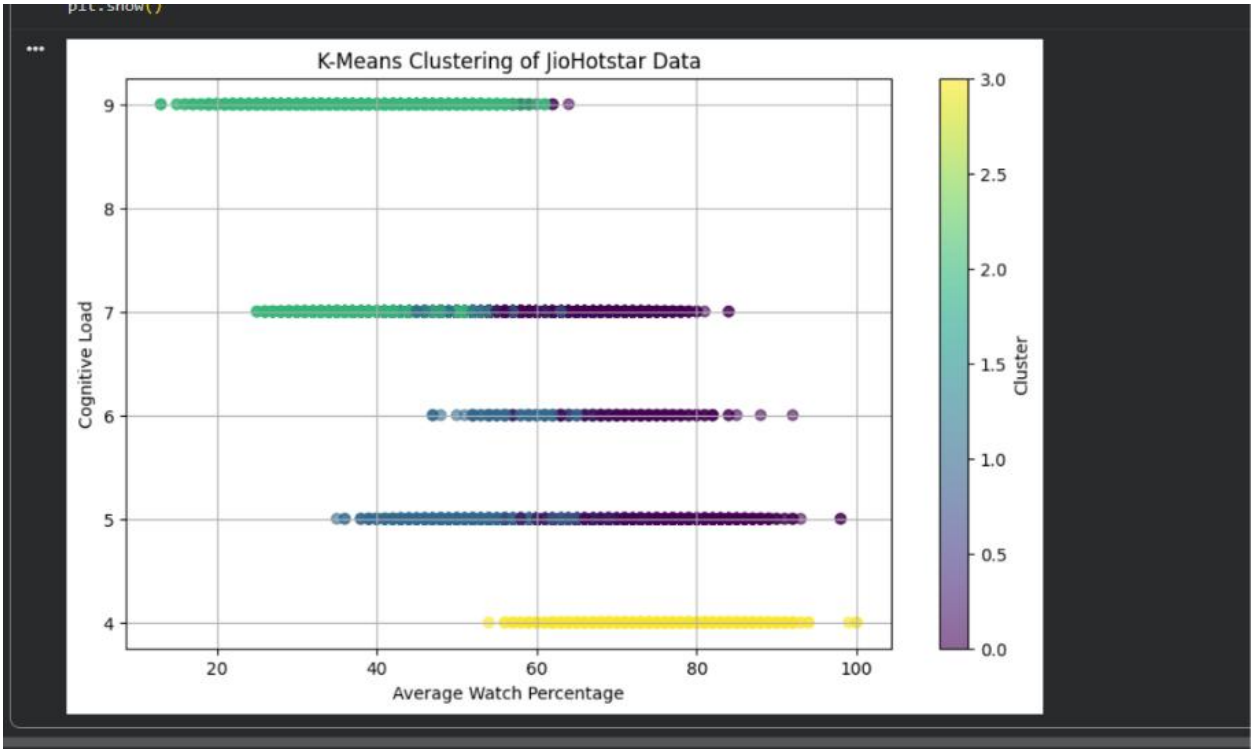
plt.figure(figsize=(10, 6))

plt.scatter(
    df['avg_watch_percentage'],
    df['cognitive_load'],
    c=df['Cluster'],
    cmap='viridis',
    alpha=0.6
)

plt.xlabel("Average Watch Percentage")
plt.ylabel("Cognitive Load")
plt.title("K-Means Clustering of JioHotstar Data")

plt.colorbar(label="Cluster")
plt.grid(True)
plt.show()
```





Inference: A scatter plot was created using average watch percentage on the X-axis and cognitive load on the Y-axis. Different colours represent the four clusters formed by K-Means. The visualization helps compare viewer-engagement patterns and observe how records are distributed among the clusters.

```
# Experiment 6 - Analyze Cluster Characteristics

cluster_summary = df.groupby('Cluster')[features].mean()

print("Mean Characteristics of Each Cluster:")
display(cluster_summary.round(2))
```

Mean Characteristics of Each Cluster:

Cluster	episode_duration_min	pacing_score	hook_strength	visual_intensity	avg_watch_percentage	pause_count	rewind_count	skip_intro	cognitive_load	night_watch_safe
0	37.23	5.75	7.09	5.97	66.44	2.81	1.98	0.00	5.79	0.0
1	37.26	6.03	3.99	6.03	56.00	2.68	2.01	1.00	5.54	0.0
2	38.22	3.99	5.05	6.05	42.59	4.08	2.04	0.62	7.69	0.0
3	46.81	8.00	5.55	5.98	75.74	2.07	1.98	0.47	4.00	1.0

Inference: Cluster-wise averages were calculated to understand the characteristics of each group. Cluster 0 showed good engagement, Cluster 1 showed moderate engagement with intro skipping, Cluster 2 showed lower watch percentage and higher cognitive load, and Cluster 3 showed the highest average watch percentage with high pacing and lower cognitive load. These results demonstrate that K-Means can identify different viewing-behaviour patterns in the JioHotstar dataset.

Conclusion:

In this experiment, K-Means clustering was successfully performed on the JioHotstar dataset using Python. The Elbow Method was used to select four clusters, and the clusters were evaluated using different performance measures. The results helped identify different viewer-behaviour patterns based on watch percentage, cognitive load, pacing score, and hook strength. Thus, the experiment demonstrated how unsupervised machine learning can be used to analyze viewer engagement and understand content-viewing patterns.